1.27 Reduction to Linear Form

Worked Solutions

1)

$y = ax^b$
$\log_{10} y = \log_{10}(ax^b)$
$\log_{10} y = \log_{10}(a) + \log_{10}(x^b)$
$\log_{10} y = \log_{10}(a) + b\log_{10}(x)$ as req.

2)

$y = a \times b^x$
$\log_{10} y = \log_{10}(a \times b^x)$
$\log_{10} y = \log_{10}(a) + \log_{10}(b^x)$
$\log_{10} y = \log_{10}(a) + x\log_{10}(b)$ as req.

3)

$y = ae^{bx}$
$\ln y = \ln(ae^{bx})$
$\ln y = \ln(a) + \ln(e^{bx})$
$\ln y = \ln(a) + bx$ as req.

4)

$y = 2 \times 1.05^x$
$\log_{10} y = \log_{10}(2 \times 1.05^x)$
$\log_{10} y = \log_{10}(2) + \log_{10}(1.05^x)$
$\log_{10} y = \log_{10}(2) + x\log_{10}(1.05)$
Let $\log_{10}(1.05) = m$ and $\log_{10}(2) = c$
So that $\log_{10} y = c + mx$ which means that the graph of $\log_{10} y$ against x will be a straight line.

5)

$y = 0.5x^{1.2}$
$\ln y = \ln(0.5x^{1.2})$
$\ln y = \ln(0.5) + \ln(x^{1.2})$
$\ln y = \ln(0.5) + 1.2\ln x$ as req
Let $\ln(0.5) = c$
So that $\ln y = c + 1.2\ln x$ which means that the graph of $\ln y$ against $\ln x$ will be a straight line.

6)

$y = 0.4 \times 0.88^{-x}$
$\log_{10} y = \log_{10}(0.4 \times 0.88^{-x})$
$\log_{10} y = \log_{10}(0.4) + \log_{10}(0.88^{-x})$
$\log_{10} y = \log_{10}(0.4) - x\log_{10}(0.88)$
Let $\log_{10}(0.88) = m$ and $\log_{10}(0.4) = c$
So that $\log_{10} y = c - mx$ which means that the graph of $\log_{10} y$ against x will be a straight line.

7 a)

$\log_{10} y = 0.477 + 5\log_{10} x$
$\log_{10} y = \log_{10}(10^{0.477}) + \log_{10}(x^5)$

$\log_{10} y = \log_{10}(10^{0.477}x^5)$
$y = 10^{0.478}x^5 = 3.0x^5$.

7 b)

$\ln y = 3\ln x + 0.815$
$\ln y = \ln x^3 + \ln(e^{0.815})$
$\ln y = \ln(e^{0.815}x^3)$
$y = e^{0.815}x^3 = 2.3x^3$.

7 c)

$\log_{10} y = -1.347 - 0.89\log_{10} x$
$\log_{10} y = \log_{10}(10^{-1.347}) - \log_{10} x^{0.89}$
$\log_{10} y = \log_{10}(10^{-1.347}x^{-0.89})$
$y = 10^{-1.347}x^{-0.89} = 0.045x^{-0.89}$.

7 d)

$\ln y = 0.51\ln x - 1.15$
$\ln y = \ln x^{0.51} + \ln(e^{-1.15})$
$\ln y = \ln(e^{-1.15}x^{0.51})$
$y = e^{-1.15}x^{0.51} = 0.32x^{0.51}$.

8 a)

$\log_{10} y = 0.214x - 1.54$
$y = 10^{0.214x-1.54}$
$y = 10^{-1.54} \times 10^{0.214x} = 0.029 \times 10^{0.214x}$
$y = 0.029 \times 1.6^x$.

8 b)

$\ln y = 0.555x + 0.444$
$y = e^{0.555x+0.444}$
$y = e^{0.444} \times e^{0.555x} = 1.6 \times e^{0.555x}$
$y = 1.6 \times 1.7^x$.

8 c)

$\ln y = -0.235x + 0.159$
$y = e^{-0.235x+0.159}$
$y = e^{0.159} \times e^{-0.235x} = 1.2 \times e^{-0.235x}$
$y = 1.2 \times 0.79^x$.

8 d)

$\log_2 y = 0.895x + 2.03$
$y = 2^{0.895x+2.03}$
$y = 2^{2.03} \times 2^{0.895x} = 4.1 \times 2^{0.895x}$
$y = 4.1 \times 1.9^x$.

9 a)

$\ln y = \ln(40e^{-0.2x})$
$\ln y = \ln(40) + \ln(e^{-0.2x})$
$\ln y = 3.7 - 0.2x$.

9 b)

$y = 0.5 \times 1.04^{0.5x}$
$\ln y = \ln(0.5 \times 1.04^{0.5x})$
$\ln y = \ln(0.5) + \ln(1.04^{0.5x})$
$\ln y = \ln(0.5) + 0.5x \ln(1.04)$
$\ln y = -0.69 + 0.020x.$

9 c)

$y = 1.62 \times 2.13^{-0.2x}$
$\ln y = \ln(1.62 \times 2.13^{-0.2x})$
$\ln y = \ln(1.62) + \ln(2.13^{-0.2x})$
$\ln y = \ln(1.62) - 0.2x \ln(2.13)$
$\ln y = 0.48 - 0.15x.$

10 a)

$\log_{10} y = 3x + 0.5$
$y = 10^{3x+0.5}$
$y = 10^{0.5} \times 10^{3x} = 3.2 \times 10^{3x} = 3.2 \times 1000^{x}.$

10 b)

$\log_{10} y = 0.25x + 0.0714$
$y = 10^{0.25x+0.0714}$
$y = 10^{0.0714} \times 10^{0.25x} = 1.2 \times 10^{0.25x} = 1.2 \times 1.8^{x}.$

11 a)

$\ln y = 4x + 2$
$y = e^{4x+2} = e^{2}e^{4x} = 7.4e^{4x}.$

11 b)

$\ln y = 0.75x + 0.1$
$y = e^{0.75x+0.1} = e^{0.1}e^{0.75x} = 1.1e^{0.75x}.$

12 a)

$\log_{10} y = 4 \log_{10}(x) + 0.1$
$\log_{10} y = \log_{10}(x^{4}) + \log_{10}(10^{0.1})$
$\log_{10} y = \log_{10}(10^{0.1}x^{4})$
$y = 10^{0.1}x^{4} = 1.3x^{4}.$

12 b)

$\log_{10} y = 0.5 \log_{10}(x) + 4$
$\log_{10} y = \log_{10}(x^{0.5}) + \log_{10}(10^{4})$
$\log_{10} y = \log_{10}(10^{4}x^{0.5})$
$y = 10^{4}x^{0.5} = 10000x^{0.5}.$

13 a)

$\log_{10} y = -5 \log_{10}(x) + 0.12$
$\log_{10} y = \log_{10}(x^{-5}) + \log_{10}(10^{0.12})$
$\log_{10} y = \log_{10}(10^{0.12}x^{-5})$
$y = 10^{0.12}x^{-5} = 1.3x^{-5}.$

13 b)

$\log_{10} y = -0.4 \log_{10}(x) + 1.17$
$\log_{10} y = \log_{10}(x^{-0.4}) + \log_{10}(10^{1.17})$
$\log_{10} y = \log_{10}(10^{1.17} x^{-0.4})$
$y = 10^{1.17} x^{-0.4} = 15x^{-0.4}$.

14)

Two points: $(0, 2)$ and $(1.8, 2.9)$
$m = \frac{2.9-2}{1.8-0} = 0.5$
$\ln y = 0.5 \ln x + 2$
$\ln y = \ln x^{0.5} + \ln(e^2) = \ln(e^2 x^{0.5})$
$y = e^2 x^{0.5} = 7.4x^{0.5}$.

15)

Two points: $(0, 3)$ and $(1.6, 3.704)$
$m = \frac{3.704-3}{1.6-0} = 0.44$
$\log_{10} y = 0.44 \log_{10} x + 3$
$\log_{10} y = \log_{10} x^{0.44} + \log_{10}(10^3) = \log_{10}(10^3 x^{0.44})$
$y = 10^3 x^{0.44} = 1000x^{0.44}$.

16)

Two points: $(0, 0.5)$ and $(1.8, 2.9)$
$m = \frac{0.803-0.5}{1.01-0} = 0.3$
$\log_{10} y = 0.3x + 0.5$
$y = 10^{0.3x+0.5} = 10^{0.5} \times 10^{0.3x}$
$y = 3.2 \times 10^{0.3x} = 3.2 \times 2.0^x$.

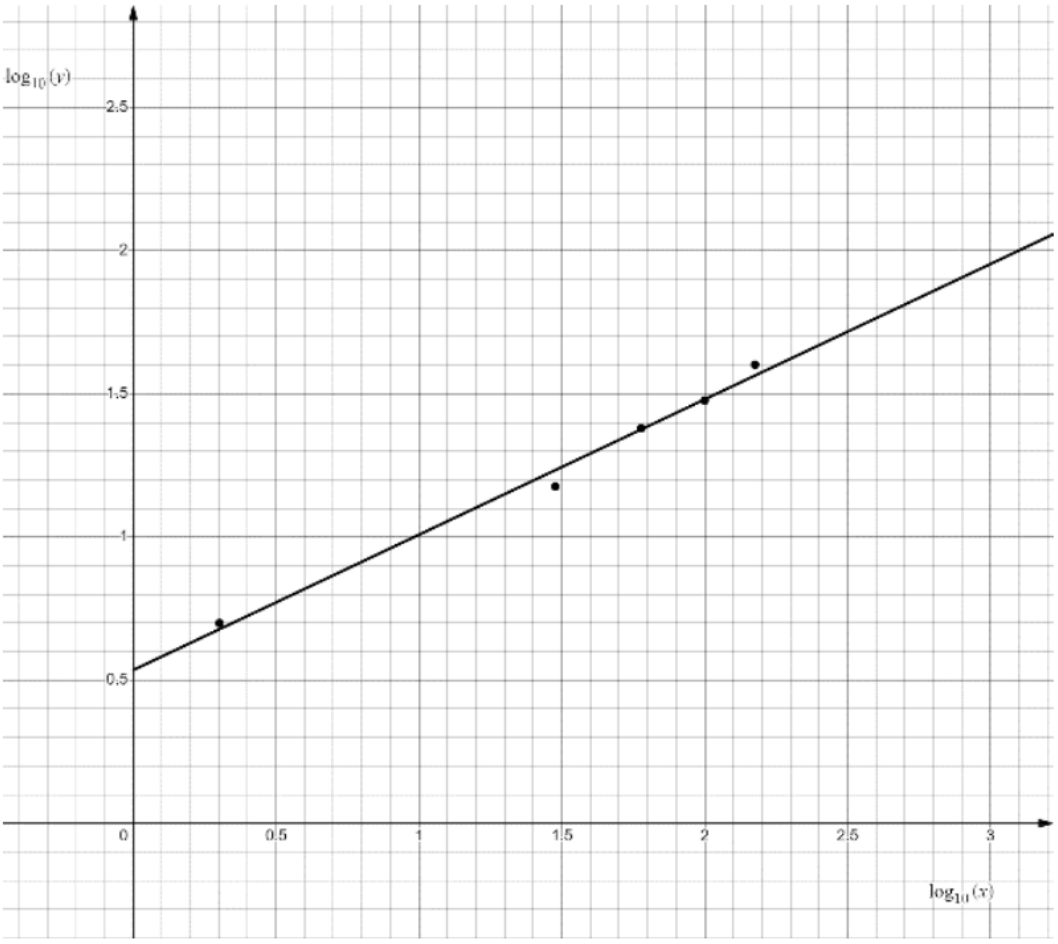
17)

Two points: $(1.1, 3.06)$ and $(2.4, 4.12)$
$m = \frac{4.12-3.06}{2.4-1.1} = \frac{53}{65}$
$y = mx + c \Rightarrow 3.06 = \frac{53}{65} \times 1.1 + c \Rightarrow c = \frac{703}{325}$
$\ln y = \frac{53}{65}x + \frac{703}{325}$
$y = e^{\frac{53}{65}x + \frac{703}{325}} = e^{\frac{703}{325}} \times e^{\frac{53}{65}x}$
$y = 8.70 \times 2.26^x$.

18 a)

$\log_{10} x$	$\log_{10} y$
0.301	0.699
1.477	1.176
1.778	1.380
2	1.477
2.176	1.602

18 b)



18 c)

y-intercept ≈ 0.54, Gradient $\approx \frac{1.38-0.54}{1.78-0} = 0.47$.

18 d)

$y = ax^b$
$\log_{10} y = \log_{10}(ax^b)$
$\log_{10} y = \log_{10}(a) + \log_{10}(x^b)$
$\log_{10} y = \log_{10} a + b \log_{10} x$ as req.

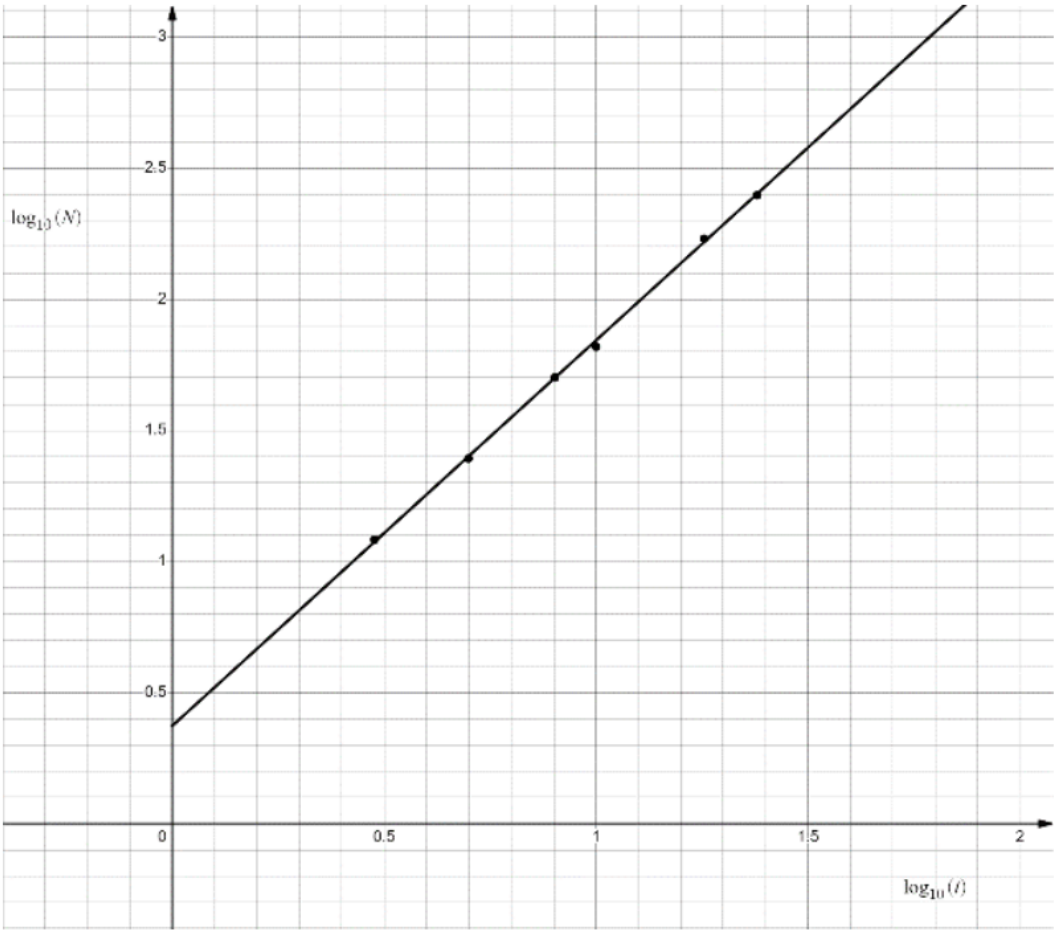
18 e)

$\log_{10} a = 0.54 \Rightarrow a = 10^{0.54} = 3.5$
$b = 0.5$.

19 a)

$\log_{10} t$	$\log_{10} N$
0.477	1.083
0.699	1.393
0.903	1.702
1	1.819
1.255	2.331
1.380	2.396

19 b)



19 c)

y-intercept ≈ 0.37, Gradient $\approx \frac{1.7-0.37}{0.9-0} = 1.47$
$\log_{10} N = \log_{10} a + b \log_{10} t$
$\log_{10} a = 0.37 \Rightarrow a = 10^{0.37} = 2.3$
$b = 1.5$.

19 d)

$N = 2.3 \times 12^{1.5} = 95.6$ million downloads.

19 e)

$200 = 2.3 \times t^{1.5} \Rightarrow t = (\frac{200}{2.3})^{\frac{1}{1.5}} = 16.4$ months.

20 a)

$\ln m$
3.912
3.040
2.186
1.281
0.531

20 b)



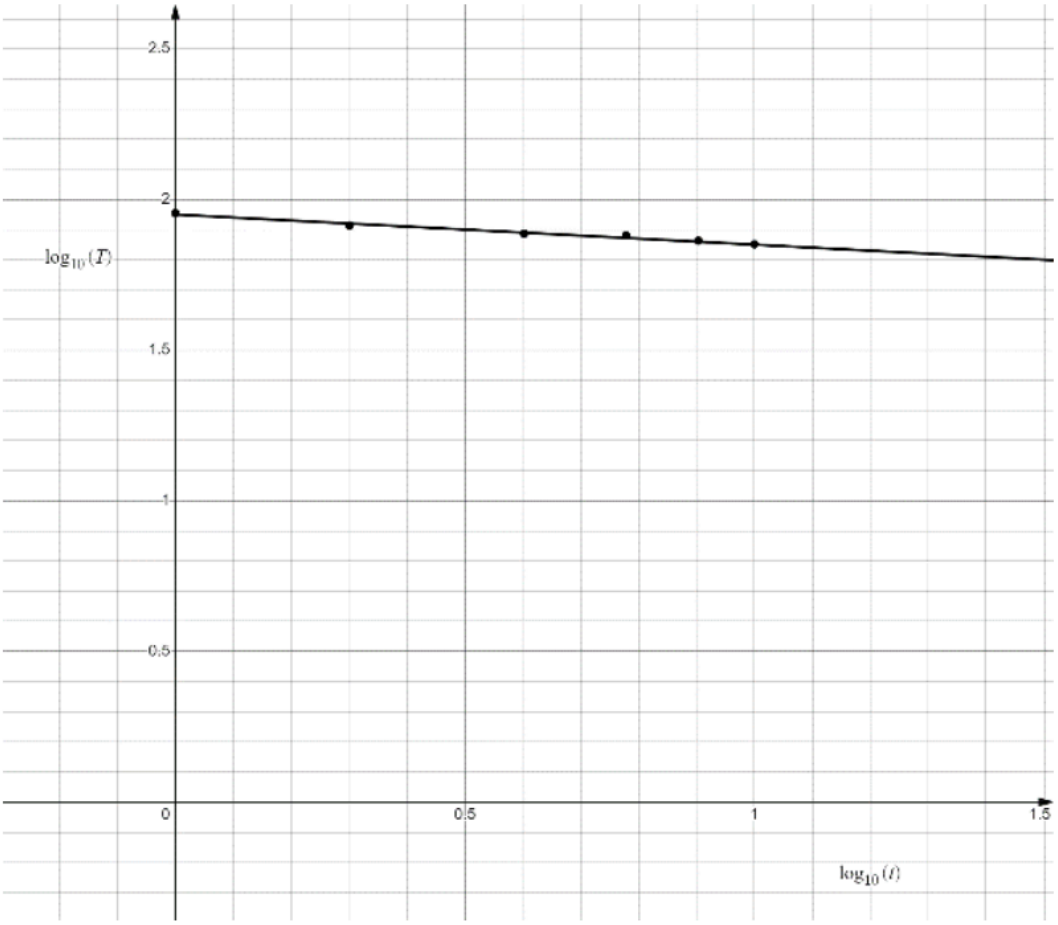
20 c)

Intercept ≈ 3.9, Gradient $\approx \frac{2.186-3.9}{20-0} = -0.0857$
$m = ae^{bt} \Rightarrow \ln m = \ln a + bt$
$\ln a = 3.9 \Rightarrow a = e^{3.9} = 49.4$
$b = -0.0857$.

20 d)

$\ln m = 3.9 - 0.0857 \times 50 \Rightarrow m = 0.68$ grams.

21 a)



21 b)

y-intercept ≈ 1.95, Gradient $\approx \frac{1.85-1.95}{1-0} = -0.1$
$\log_{10} T = \log_{10} a + b \log_{10} t$
$\log_{10} a = 1.95 \Rightarrow a = 10^{1.95} = 89$
$b = -0.1$
$T = 89 \times t^{-0.1}$.

21 c)

a is the initial amount of time spent exercising (that is when 0 weeks have passed)

21 d)

$T = 89 \times 3^{-0.1} = 80$ minutes.

21 e)

When $t = 0$ there are no solutions so the model cannot be used to predict the amount of exercise done at this point.

21 f)

Morgana may well continue to exercise regularly as she is still doing 10 weeks in, and so has formed a habit. But it is unlikely it will continue to decrease in this manner consistently for a long period of time.

22 a)

Since $y = ax^b \Rightarrow \log_{10} y = \log_{10} a + b \log_{10} x$
He should have plotted $\log_{10} y$ against $\log_{10} x$.

22 b)

$\log_{10} y = 2.4 + 0.6x$
$y = 10^{2.4+0.6x} = 10^{2.4} \times 10^{0.6x} = 251 \times 4^x$.

23 a)

She raised the right-hand side of her equation by 10 to the power rather than e.

23 b)

Gradient is -0.0395, y-intercept is 7.82
$y = ae^{bx} \Rightarrow \ln y = \ln a + bx$
$\ln y = -0.0395x + 7.82$
$y = e^{-0.0395x+7.82}$.

24 a)

$\ln N = 7 + \frac{4.6-7}{3-0} \ln t$
$\ln N = 7 - 0.8 \ln t$.

24 b)

$\ln N = \ln(e^7) + \ln(t^{-0.8})$
$\ln N = \ln(e^7 t^{-0.8})$
$N = 1096.6t^{-0.8}$.

24 c)

$t = 66$
$N = 1096.6 \times 66^{-0.8} = 38.4$ million cinema admissions.

24 d)

Due to the pandemic, there was a huge drop in cinema attendance in reality.

25 a)

Gradient: $\frac{24.86-20.19}{25-15} = 0.467$
$20.19 = 0.467 \times 15 + c \Rightarrow c = 13.185$
$\log_2 y = 0.467x + 13.185$.

25 b)

$y = 2^{0.467x+13.185} = 2^{13.185} 2^{0.467x} = 9312.8 \times 2^{0.467x}$.

25 c)

a is the number of transistors in a microprocessor in 1975, according to the model.

25 d)

$x = 55$
$9312.8 \times 2^{0.467 \times 55} = 5.0 \times 10^{11}$ transistors.

26 a)

$\log_{10} N = 0.238t + 1.975$
$N = 10^{0.238t+1.975} = 10^{1.975} \times 10^{0.238t}$
$= 99.41 \times 1.73^t$.

26 b)

a is the initial number of subscribers according to the model.
b is the exponential rate, ie the number of subscribers is increasing by 73% per year.

26 c)

$\log_{10} N = 0.238 \times 10 + 1.975$
$N = 10^{4.355} = 22646$ subscribers.

27 a)

$t = 0$
$\ln T = 4.585 \Rightarrow T = e^{4.585} = 98.0°\text{C}$.

27 b)

$T = 98.0e^{-0.0701t}$.

28 a)

$\ln T = \ln(d^{1.48}) - \ln(e^{1.6})$
$\ln T = \ln(e^{-1.6}d^{1.48})$
$T = e^{-1.6}d^{1.48}$
$T = 0.203d^{1.48}$.

28 b)

$T = 0.203 \times 228^{1.48} = 624$ days.

28 c)

$\frac{687-624}{687} \times 100 = 9.17\%$.

29 a)

$r = a \times T^b$
$\ln r = \ln(a \times T^b)$
$\ln r = \ln(a) + \ln(T^b)$
$\ln r = \ln(a) + b\ln(T)$ as req.

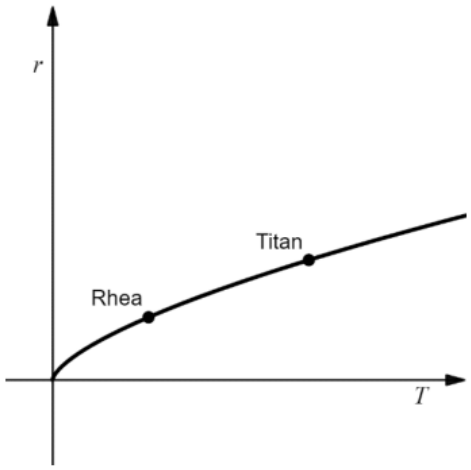
29 b)

$\ln r$	$\ln T$
0.199	2.766
-0.635	1.504

Gradient: $b = \frac{0.199+0.635}{2.766-1.504} = \frac{417}{631} = 0.66$
y-intercept:
$0.199 = \frac{417}{631} \times 2.766 + c \Rightarrow c = -1.63$
$a = \ln(-1.63) \Rightarrow a = e^{-1.63} = 0.196$
$\ln r = 0.196 + 0.66 \ln T$.

29 c)

$r = 0.196T^{0.66}$



29 d)

$r = 0.196 \times 2.7^{0.66} = 0.378$ million km.

29 e)

$\frac{0.38-0.378}{0.38} \times 100 = 0.53\%$.

30 a)

$\ln 71 = \ln b - 5a$
$\ln 24 = \ln b - 55a$
$\ln 71 - \ln 24 = -5a + 50a \Rightarrow a = 0.0217$.

30 b)

$\ln 71 = \ln b - 5 \times 0.0217\ldots$
$\ln 71 + 5 \times 0.0217\ldots = \ln b$
$b = e^{\ln 71 + 5 \times 0.0217\ldots} = 79.1$.

30 c)

$\ln y = 4.37 - 0.0217x$
$\ln y = 4.37 - 0.0217 \times 40$
$y = e^{4.37-0.0217\times 40} = 33.2$ years
~ 33 years.

30 d)

$\ln 25 = 4.37 - 0.0217x$
$x = 53.1$
~ 53 years.

31)

$m = \frac{8.49-5.41}{4.7-0.3} = 0.7$
$8.49 = 0.7 \times 4.7 + c \Rightarrow y$-intercept is 5.2
$y = 0.7x^2 + 5.2$.

32)

$m = \frac{7.8-5.3}{\frac{3\pi}{5}-\frac{\pi}{2}} = \frac{\frac{5}{2}}{\frac{\pi}{10}} = \frac{25}{\pi}$
$5.3 = \frac{25}{\pi} \times \frac{\pi}{2} + c \Rightarrow y$-intercept is $-\frac{36}{5}$
$y = \frac{25}{\pi}\sin(x) - \frac{36}{5}$.